# Jiali He 何佳俐

96 Jinzhai Road, Hefei, China, 230026

+86 17376549348 | hjl542542@mail.ustc.edu.cn | Personal Website | GitHub

## EDUCATION

**University of Science and Technology of China (USTC)** Hefei, China

*B.S. in Statistics, Department of Statistics and Finance, School of Management* 09/2023 – 06/2027 (Expected)

- GPA: **3.93/4.30** Weighted Average: **90.97** Rank: **1/66** in School of Management
- Language Proficiency:
  TOEFL iBT **6.0** [**118 of 120**]: R6.0/L6.0/**S6.0**/W5.5
  GRE **330**: V160/Q170/AW4.0
- Programming Skills: Skilled in R, Python, C/C++; Familiar with MATLAB, SPSS.
- Scholarships:
  China National Scholarship (Top 3% in school) 09/2026
  Chen Linyi Scholarship (Top 5% in school) 09/2025
  China National Scholarship (Top 3% in school) 09/2024
- Honors:
  Finalist, Mathematical Contest in Modeling (MCM) 05/2026
  Second Prize, Chinese Mathematics Competitions 12/2024

## RESEARCH INTERESTS

Computational Statistics, Bayesian Statistics, Statistical Machine Learning

## PUBLICATION

1. **He, J.** and Liu, S. (2026). *Pseudo-marginal Sampling for Selective Inference.* arXiv:2611.13492 · Code

## RESEARCH EXPERIENCE

**Pseudo-marginal Sampling for Selective Inference** 04/2026 – Present

*Advisor: Prof. Sifan Liu (Department of Statistical Science, Duke University)*

- Implemented independent and correlated pseudo-marginal MCMC samplers that use simulated selection outcomes for selective inference with analytically intractable selection probabilities.
- Designed and ran all simulations to assess sampling efficiency, Type I error, and power in a Gaussian file-drawer benchmark, cross-validated lasso, and randomized changepoint detection.
- Measured several-hundred-fold gains in sampling efficiency over rejection sampling in some settings.
- Drafted the manuscript and checked proofs of cost bounds and sample complexity results.

**Prediction-Oriented Multivariate Shuffled Linear Regression** 09/2026 – Present

*Advisor: Prof. Qing Yang (Department of Statistics and Finance, USTC)*

- Proposed studying how residual predictor–response mismatches affect out-of-sample prediction risk.
- Derived risk decompositions and proved sufficient conditions for consistency and first-order oracle equivalence.
- Designed controlled experiments and wrote simulation code to test the theoretical predictions.
- Extending the analysis to high-dimensional settings.

**Return Subsidies and Quality Signaling in Livestream Commerce** 10/2025 – 05/2026

*Project Lead, National Undergraduate Innovation Training Program*

*Advisor: Prof. Jie Wu (Department of Management Science, USTC); Postdoctoral Collaborator: Xiongfei Guo*

- Formulated a Bayesian signaling game linking return subsidies to consumers' beliefs about product quality.
- Derived purchase and return thresholds and proved the existence of a least-cost separating equilibrium.
- Documented the findings in a complete research manuscript, which was incorporated into the team's final report.
- Led a three-person team, coordinated meetings and progress, and presented at the final defense; the project received RMB 20,000 in funding and an Excellent rating.

## PROJECT EXPERIENCE

**Seasonal Revenue Forecasting for Online Retail** 06/2026

*Time Series Analysis Course Project, USTC*

- Cleaned 1.07 million transaction records, built a daily revenue series, and fitted seasonal forecasting models.
- Compared forecasts using a rolling holdout and rolling-origin cross-validation. SARIMA had the lowest MAE and RMSE in the year-end holdout.

**Latent Vote Inference and Scoring Rule Evaluation** 01/2026

*MCM Finalist, Team Lead Advisor: Prof. Jing Zeng (Department of Statistics and Finance, USTC)*

- Checked vote reconstructions against historical eliminations and quantified posterior uncertainty.
- Compared scoring rules and proposed a hybrid rule combining percentage scoring with a judges' save.
- Led a three-person team by assigning tasks, running progress meetings, and synthesizing interim results.

## RELEVANT COURSES

*(OG denotes ongoing courses; * denotes graduate-level courses)*

- Mathematics: Mathematical Analysis B2 (99), Linear Algebra B1 (90), Introduction to Differential Equations with Applications (96), Real Analysis (85), Functional Analysis (85).
- Statistics: Mathematical Statistics (95), Applied Stochastic Processes (95), Regression Analysis (96), Multivariate Analysis (96), Time Series Analysis (95), Non-parametric Statistics (95), Fundamentals of Statistical Algorithm (92), *Advanced Probability (OG), *Bayesian Analysis (OG), Machine Learning (OG).
- Computer Science: Computer Programming (94), Applied Statistical Software (92).

## ACADEMIC PROGRAMS

**International Summer Session, Fudan University** Shanghai, China

*One of five students selected by USTC* 06/2025 – 07/2025

- Courses: Introduction to Python ($A^+$), Artificial Intelligence in FinTech ($A^+$).

**Tongali Entrepreneurship Program, Nagoya University** Nagoya, Japan

*Team Lead and Presenter* 02/2025

- Visited Mirai Project's adult day care center in Nagoya and spoke one-on-one with a center representative about local elder care needs and our proposal.
- Co-developed a proposal for IoT health monitoring and AI risk alerts in a three-person team.

## LEADERSHIP AND ACTIVITIES

**USTC Student Dance Troupe** 09/2023 – Present

*Leadership Team Member (two years); Publicity Coordinator; Performer*

- Led publicity, helped organize two annual dance productions, and performed in 10+ campus events.

**USTC Aerobics Club** 09/2024 – Present

*Member and Performer*

- Performed in 10+ campus events; represented USTC in the 2024 Anhui Provincial University Dragon and Lion Dance Competition, earning a team Second Prize (3rd place) in Ribbon Dragon Dance.